Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
MERN Stack

Practical 8

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Class: TYIT	Batch: 1
Date of Assignment: 23/9/2026	Date/Time of Submission: 23/9/2026

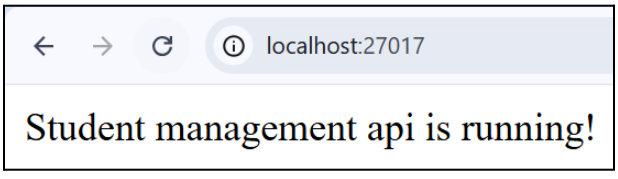
a. Create a simple REST API using Express
Steps:
1) Create a folder & open that in vs code
2) Run -> npm init -y to create package.json file.
3) Add server.js file
4) Run command -> npm install mongoose express cors
5) Add following code into server.js file
6) Type cmd: node server.js to test, type: localhost: 27017 in browser to get output

Code:

```
const express = require('express');
const mongoose = require('mongoose');
const cors = require('cors'); const PORT = 27017;
app = express();app.use(cors());app.use(express.json());
mongoose.connect("mongodb://localhost:27017/CollegeDB")
.then(()=>{ console.log("Mongodb Connected with Node.js");})
.catch((error)=>{ console.log("An error occured: ",error);})
app.get("/",(req,res)=>{
    res.send("Student management api is running!");})
app.listen(PORT,()=>{
    console.log(`Server is running on mongodb://localhost:${PORT}`);});
```

Output:

```
PS C:\Users\Hanoch\Desktop\College TY\Sem 5\MERN\practical 8> node server.js
Server is running on mongodb://localhost:27017
```



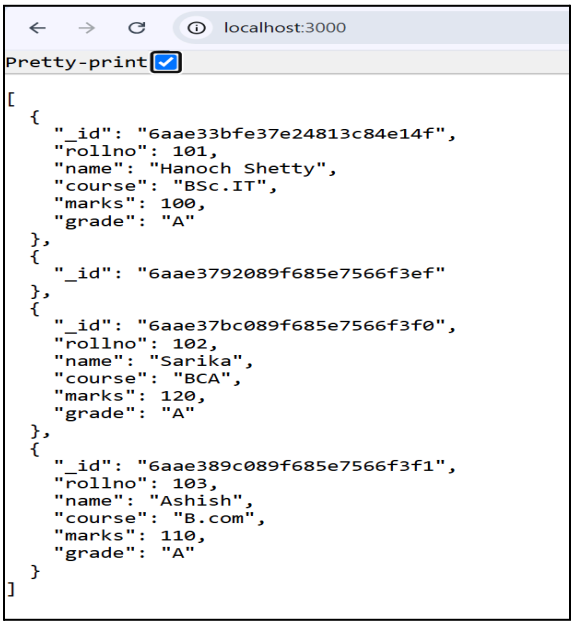
b. Implement the GET API to retrieve data

Steps: Change the app.get() code rest everything is as same as in 8a

Code:

```
const express = require('express');
const mongoose = require('mongoose');
const cors = require('cors');
const PORT = 3000;
app = express();app.use(cors());
app.use(express.json());
mongoose.connect("mongodb://localhost:27017/CollegeDB")
.then(()=>{
    console.log("Mongodb Connected with Node.js");})
.catch((error)=>{
    console.log("An error occured: ",error);})
app.get("/", async (req, res) => {
try {
const students = await
mongoose.connection.db.collection("students").find().toArray();
res.json(students);}
catch (error) {
res.status(500).json({
message: "Error fetching students"});}});
app.listen(PORT,()=>{console.log(`Server is running on localhost:${PORT}`);
});
```

Output:



C. Fetch Data from the api in a react application

Steps:

1)Create react project using following command

npm create vite@latest my-react

2)Choose react, javascript, ESLint

3)In terminal move to my-react & install axios

npm install axios

4)Add following code to App.jsx

Code;

App.jsx:

```
import {useEffect, useState} from 'react';

import axios from 'axios';


export default function App(){

  const [students,setStudents] = useState([]);


useEffect(()=>{axios.get("http://localhost:3000/Students").then((response)=>{setStud
ents(response.data);})

  .catch((error)=>{console.log(error);}); }, []);


  return(

    <>

      <h1>Student List</h1>

      <table border="2">

        <thead>

          <tr>

            <th>Roll No</th>

            <th>Name</th>

            <th>Couse</th>
```

```
            <th>Marks</th>
          </tr>
        </thead>
        <tbody>
          {students.map((students)=>(
            <tr id={students.id}>
              <td>{students.rollNo}</td>
              <td>{students.name}</td>
              <td>{students.course}</td>
              <td>{students.marks}</td>
            </tr>
          ))}
        </tbody>
      </table>
    </>
  )
}
```

server.js

```
import express from 'express';import mongoose from 'mongoose';import cors from
'cors';
const PORT = 3000
const app = express();
app.use(cors());
app.use(express.json());
mongoose.connect("mongodb://localhost:27017/CollegeDB")
```

```
.then(()=>{console.log("Mongodb Connected with Node.js");})

.catch((error)=>{ console.log("An error occured: ",error);})

app.get("/Students", async (req, res) => {

try {const students = await
mongoose.connection.db.collection("students").find().toArray();

res.json(students);}catch {

res.status(500).json({

message: "Error fetching students"});}});

app.listen(PORT,()=>{console.log(`Server is running on
mongodb://localhost:${PORT}`);});
```

Output:

```
  VITE v8.3.0  ready in 1782 ms

  → Local:   http://localhost:5173/
  → Network: use --host to expose
  → press h + enter to show help
```

```
PS C:\Users\Hanoch\Desktop\College TY\Sem 5\MERN\practical
8\practical8c> node server.js
Server is running on mongodb://localhost:3000
Mongodb Connected with Node.js
```

Student List

Roll No	Name	Couse	Marks
	Hanoch Shetty	BSc.IT	100
	Sarika	BCA	120
	Ashish	B.com	110